Visualization

2026-01-07

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

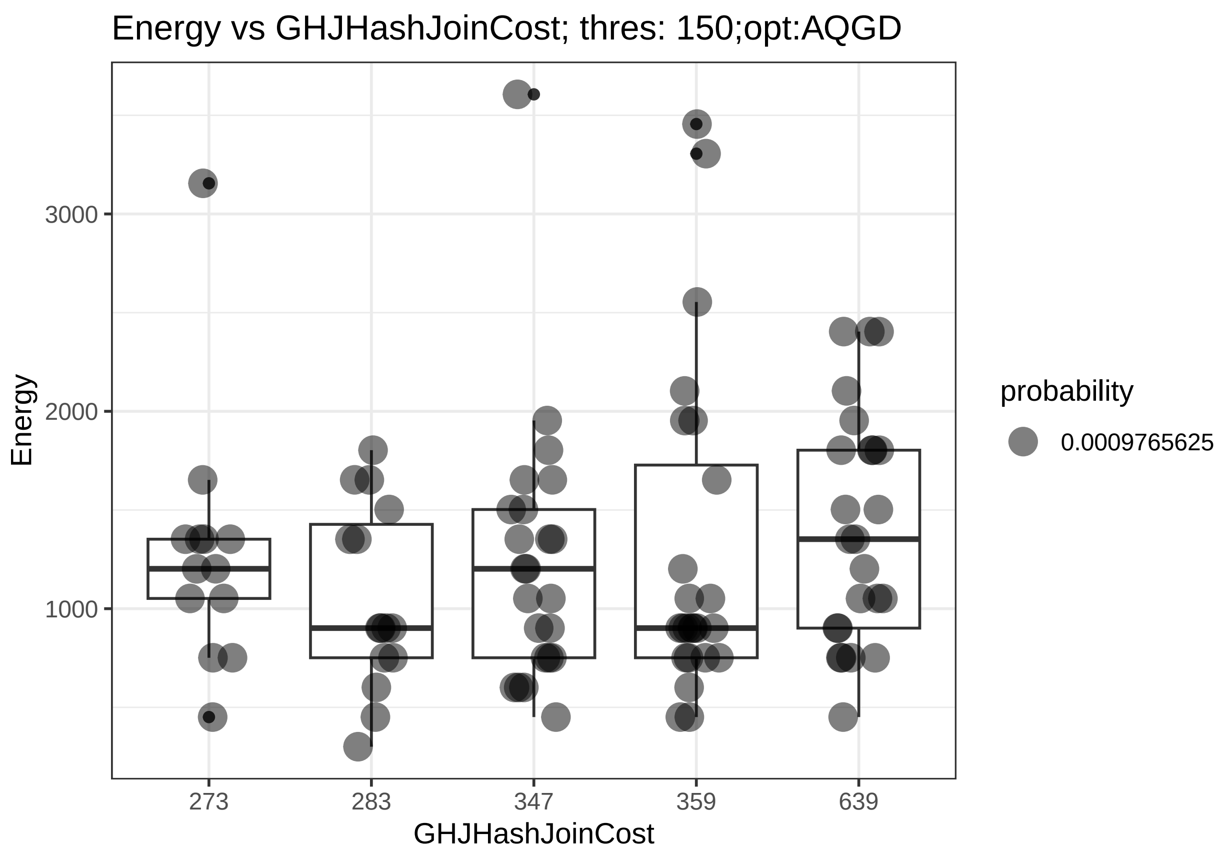
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

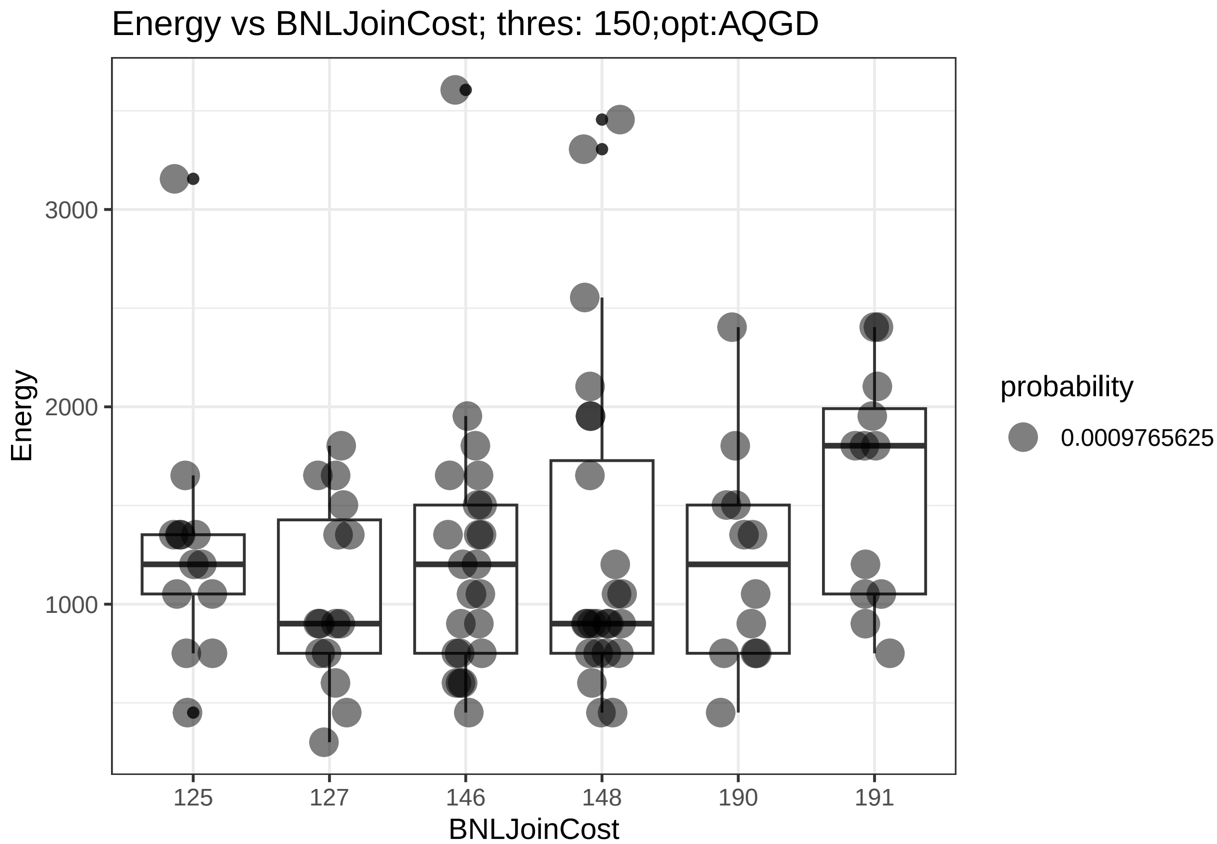
```
cost_energy_boxplot_prob_noweight <- function(file_path, show_points = TRUE,targetCol="cost",thresh="()
  library(readr)
  library(dplyr)
  library(tidyr)
  library(ggplot2)

  df <- read_csv(file_path, show_col_types = FALSE)

  if (!("probability" %in% names(df)) && ("prob" %in% names(df))) {
    df <- df %>% rename(probability = prob)
  }

  df <- df %>%
    transmute(
      cost = as.numeric(.data[[targetCol]]),
      energy = as.numeric(energy),
      probability = as.numeric(probability)
    ) %>%
    filter(is.finite(energy), is.finite(probability), probability > 0)

  w0 <- min(df$probability)
  df_exp <- df %>%
    mutate(w = pmax(1L, round(probability / w0))) %>%
    uncount(w)
```

```
  p <- ggplot(df_exp, aes(x = factor(cost), y = energy)) +
    geom_boxplot() +
    theme_bw() +
    labs(x = targetCol, y = "Energy",
         title = glue("Energy vs {targetCol}; thres: {thresh};opt:{opt} "))
  print(p)
}
```

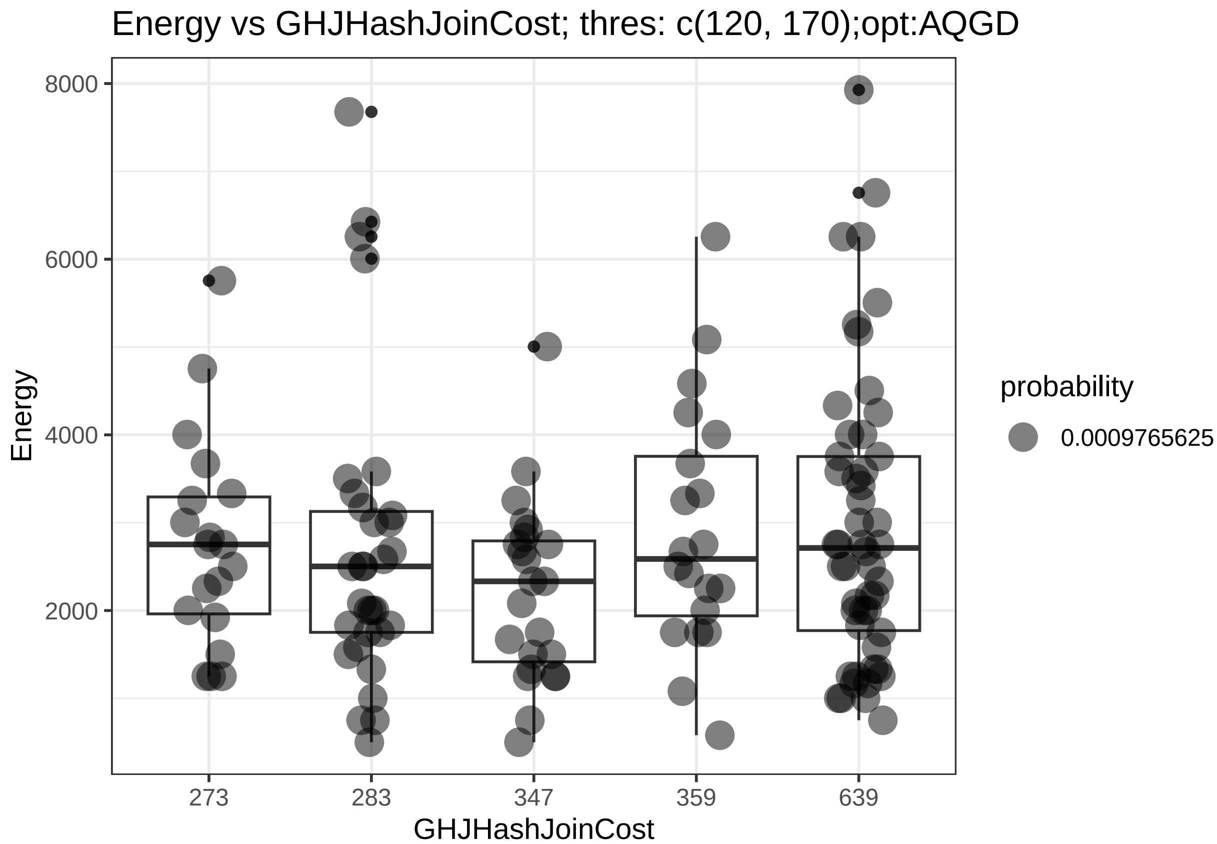
```
## Input cardi: [10,15,20]
```



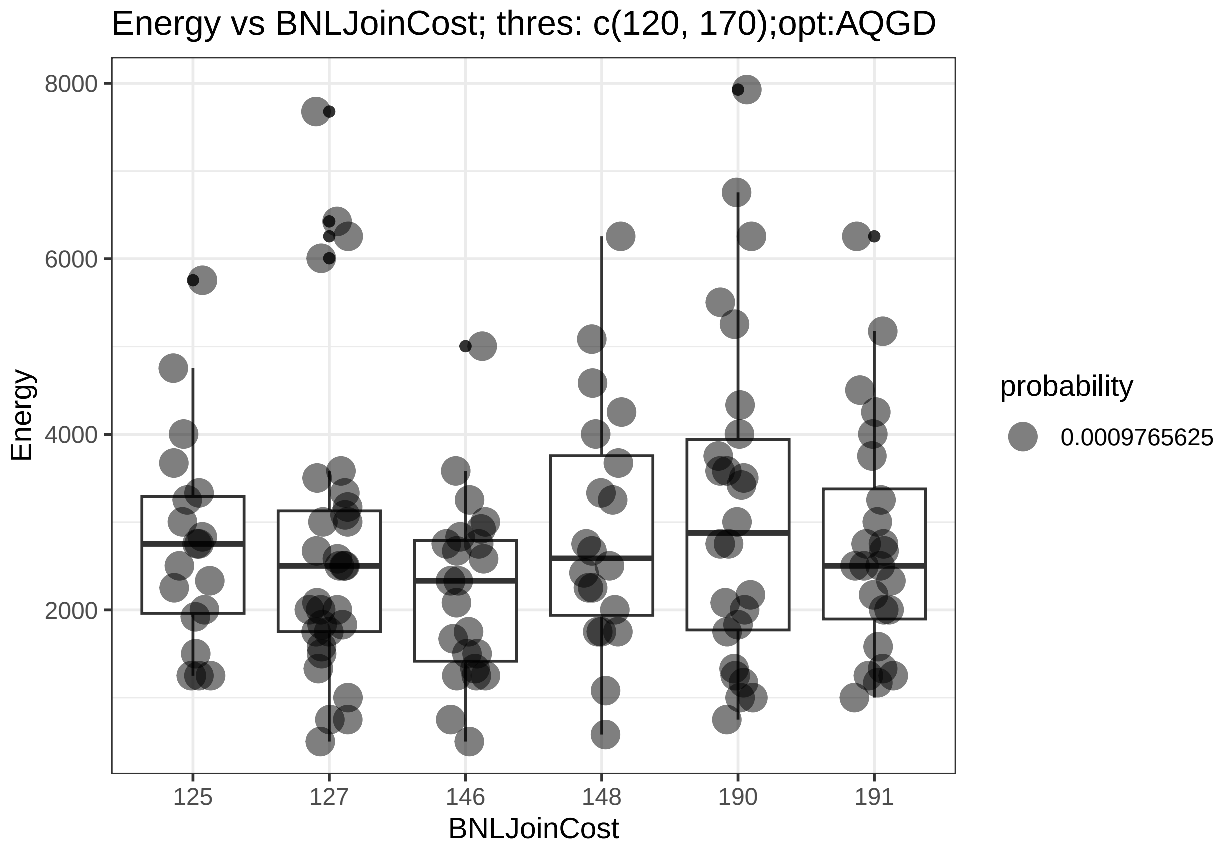

```
## optimizerAQGD; input thres:(150); costType:GHJHashJoinCost
```




```
## optimizerAQGD; input thres:(150); costType:BNLJoinCost
```




```
## optimizerAQGD; input thres:(c(120, 170)); costType:GHJHashJoinCost
```

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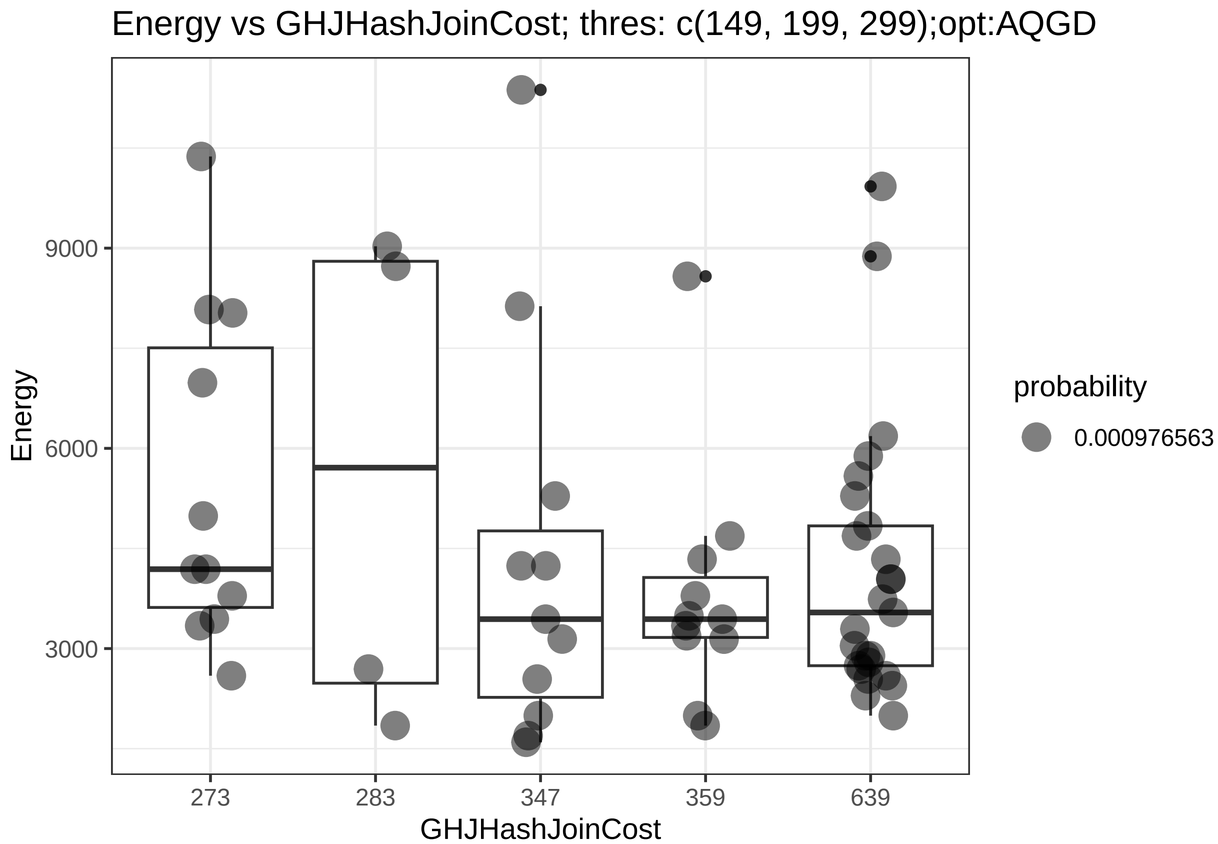
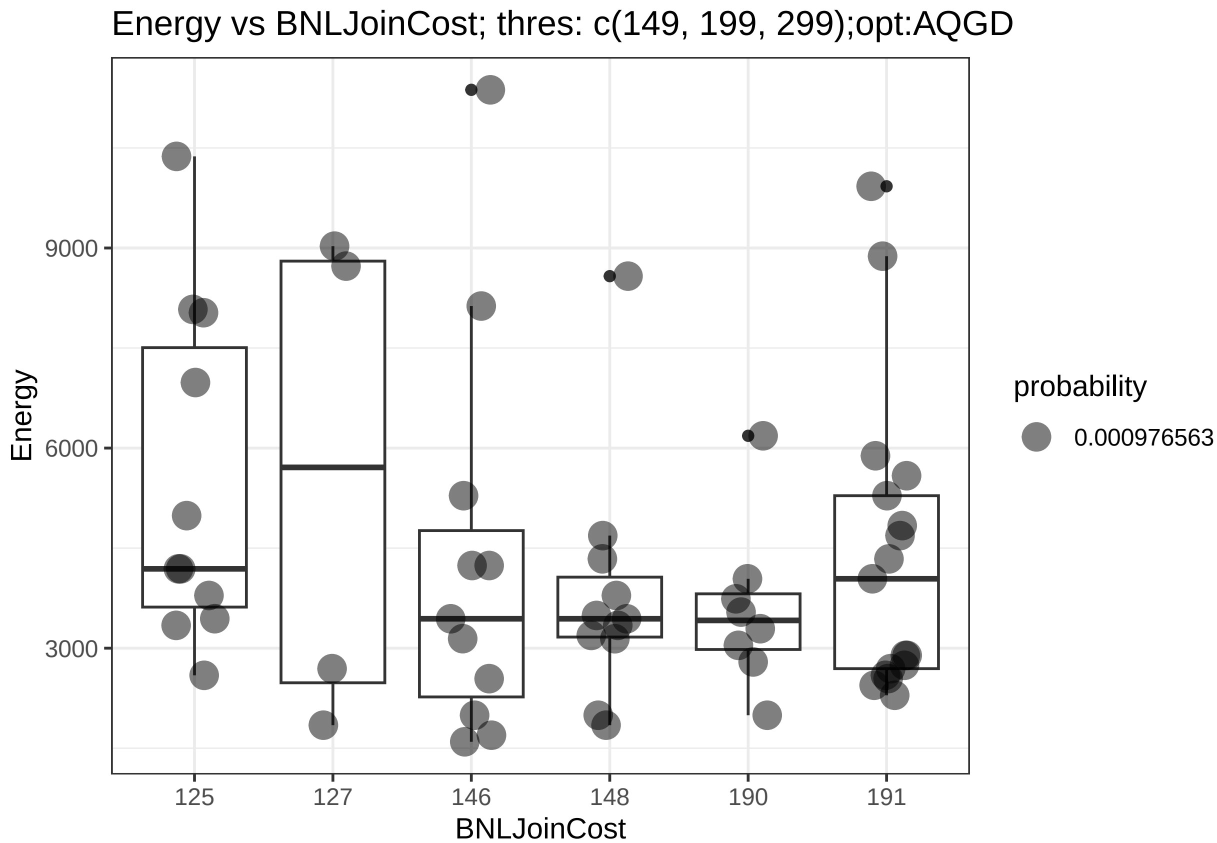



```
## optimizerAQGD; input thres:(c(149, 199, 299)); costType:GHJHashJoinCost
```

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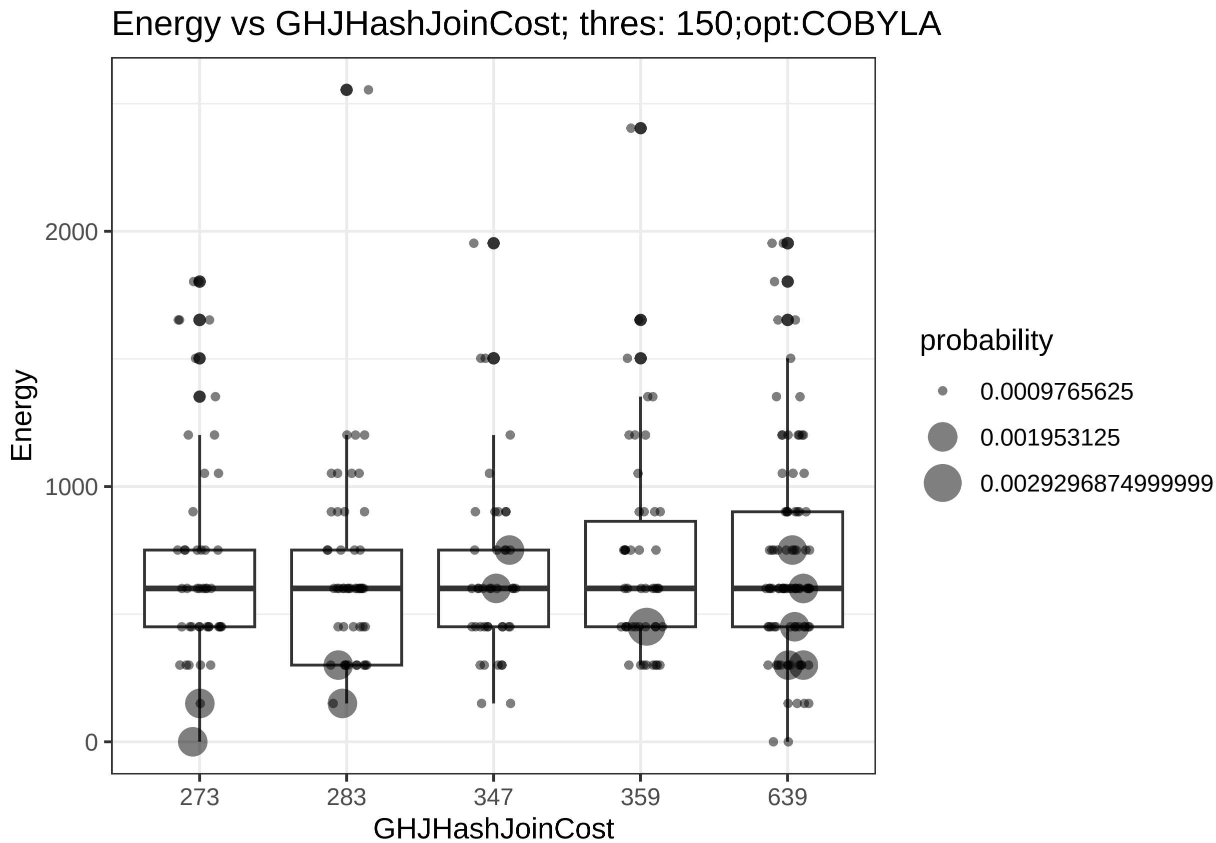
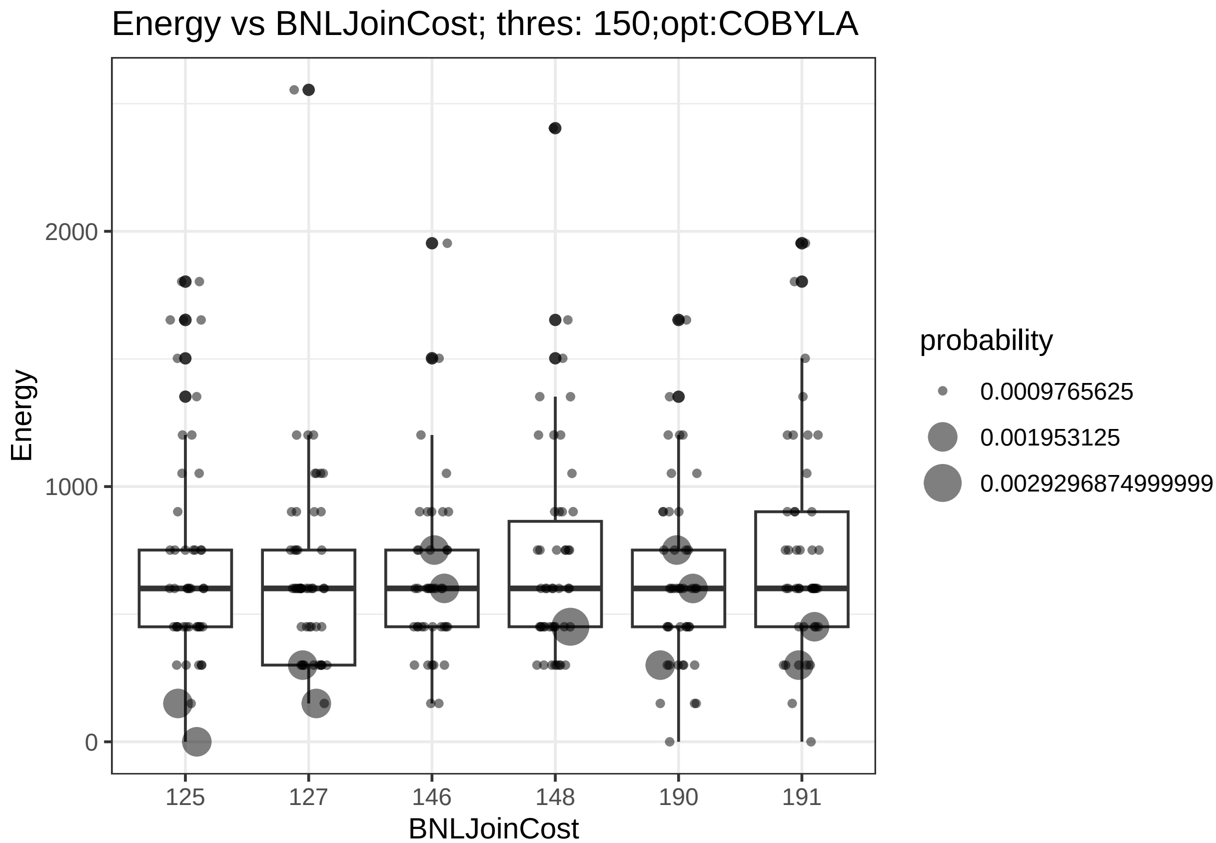



```
## optimizerCOBYLA; input thres:(150); costType:GHJHashJoinCost
```

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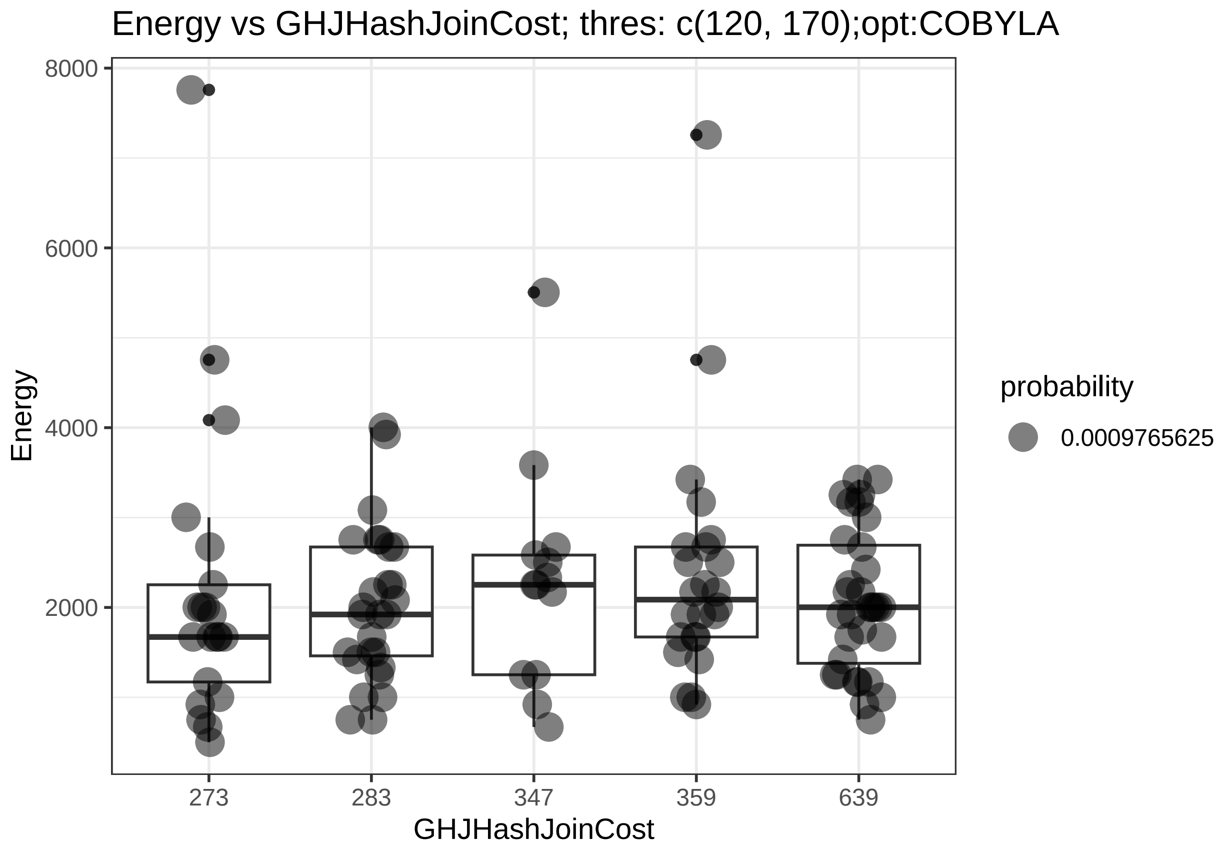


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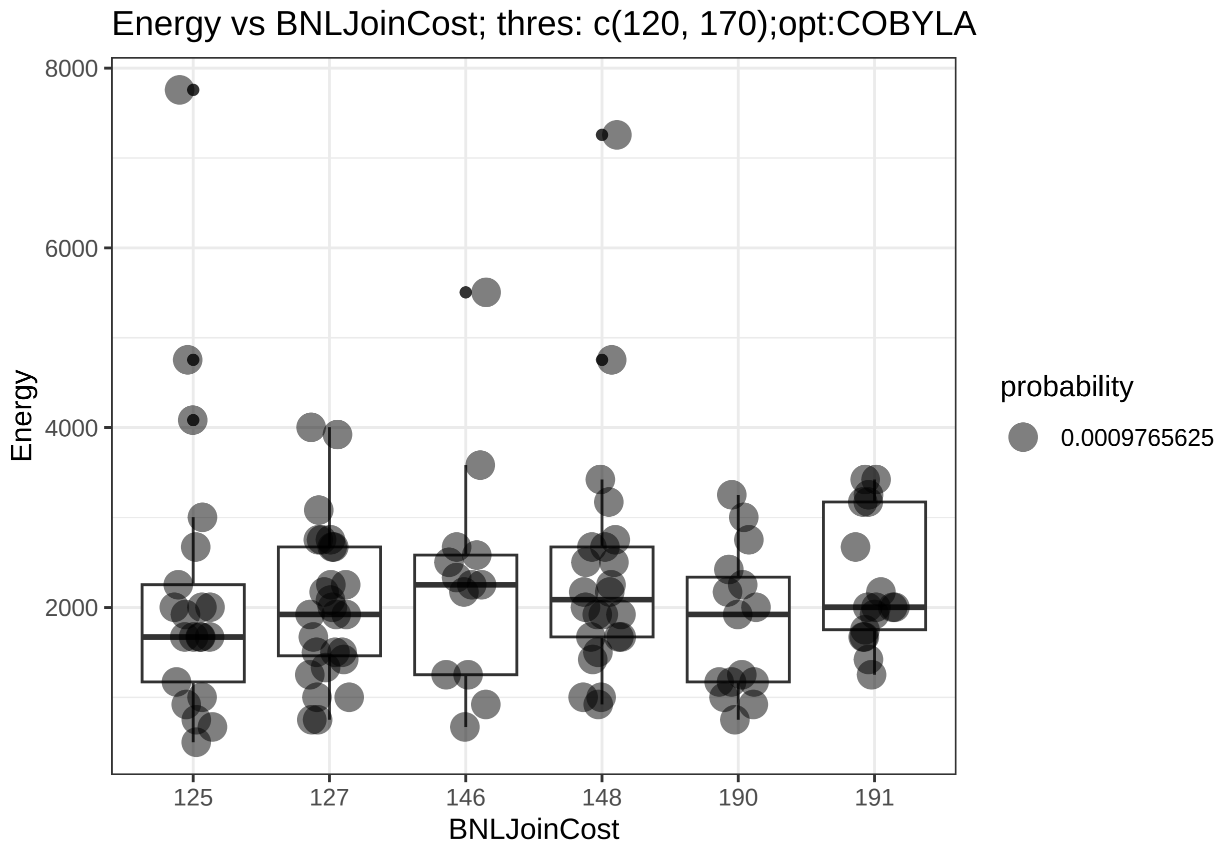


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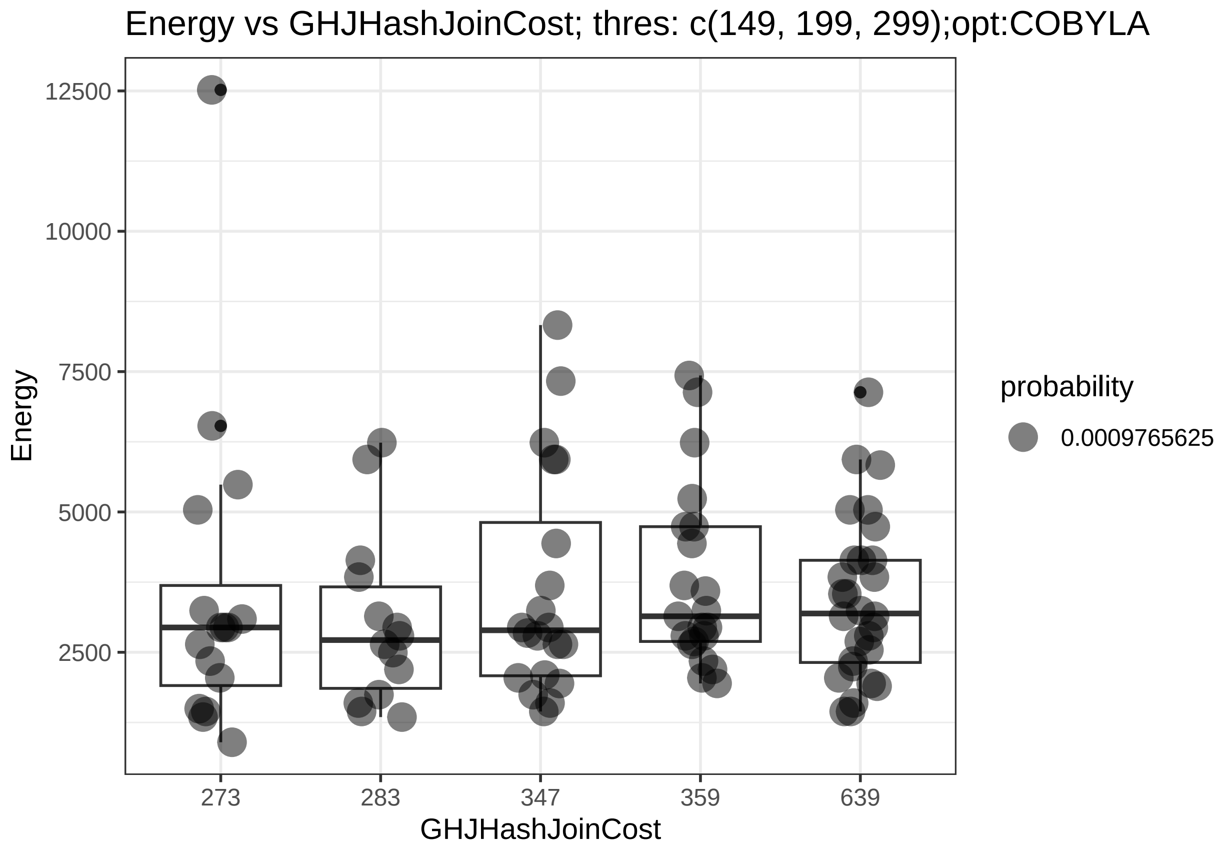


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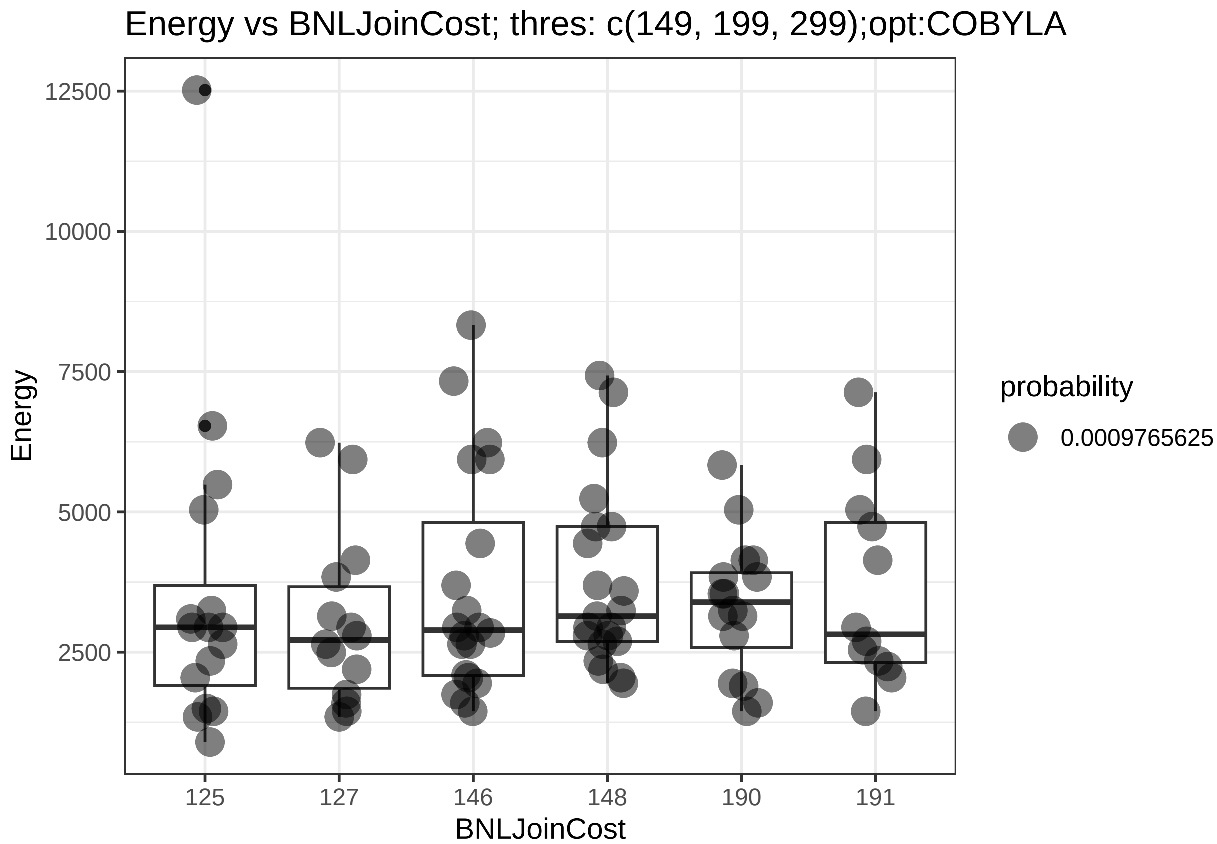


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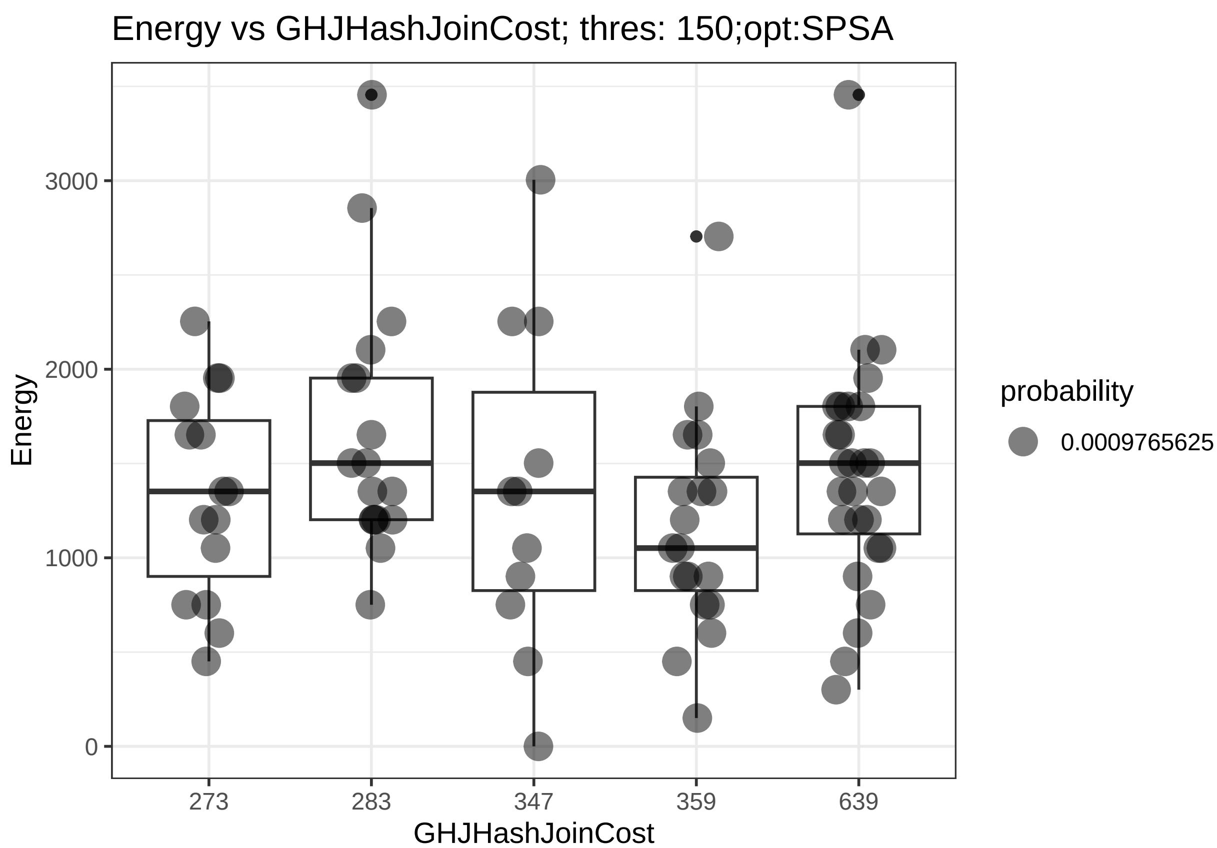


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## optimizerSPSA; input thres:(150); costType:GHJHashJoinCost
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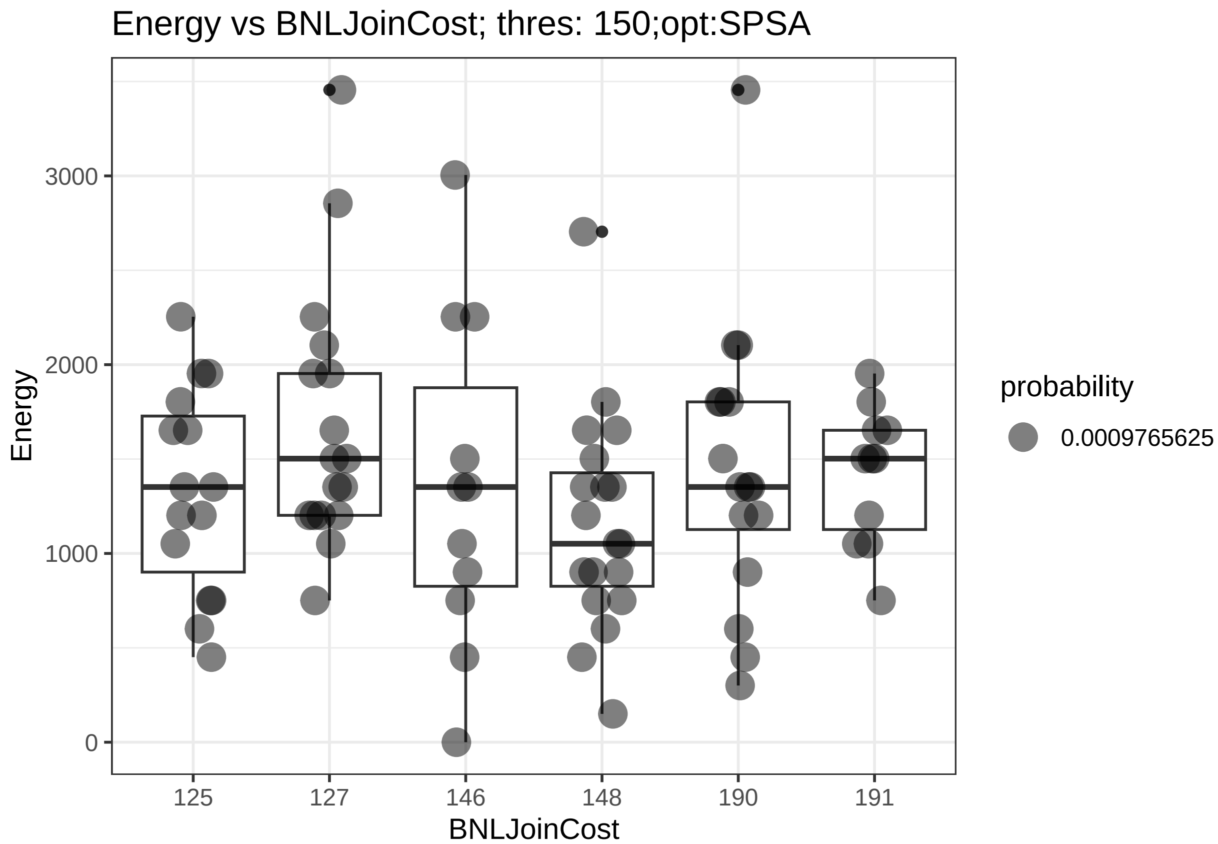


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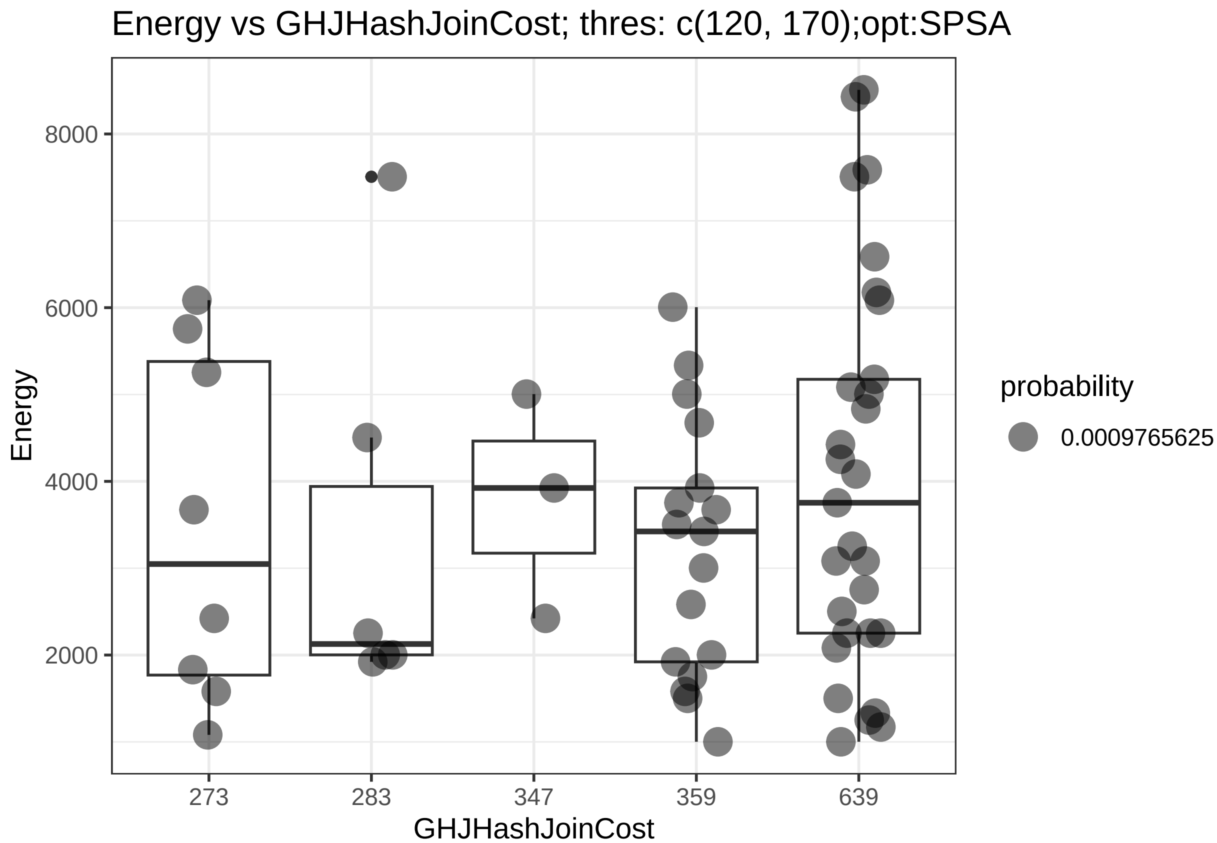


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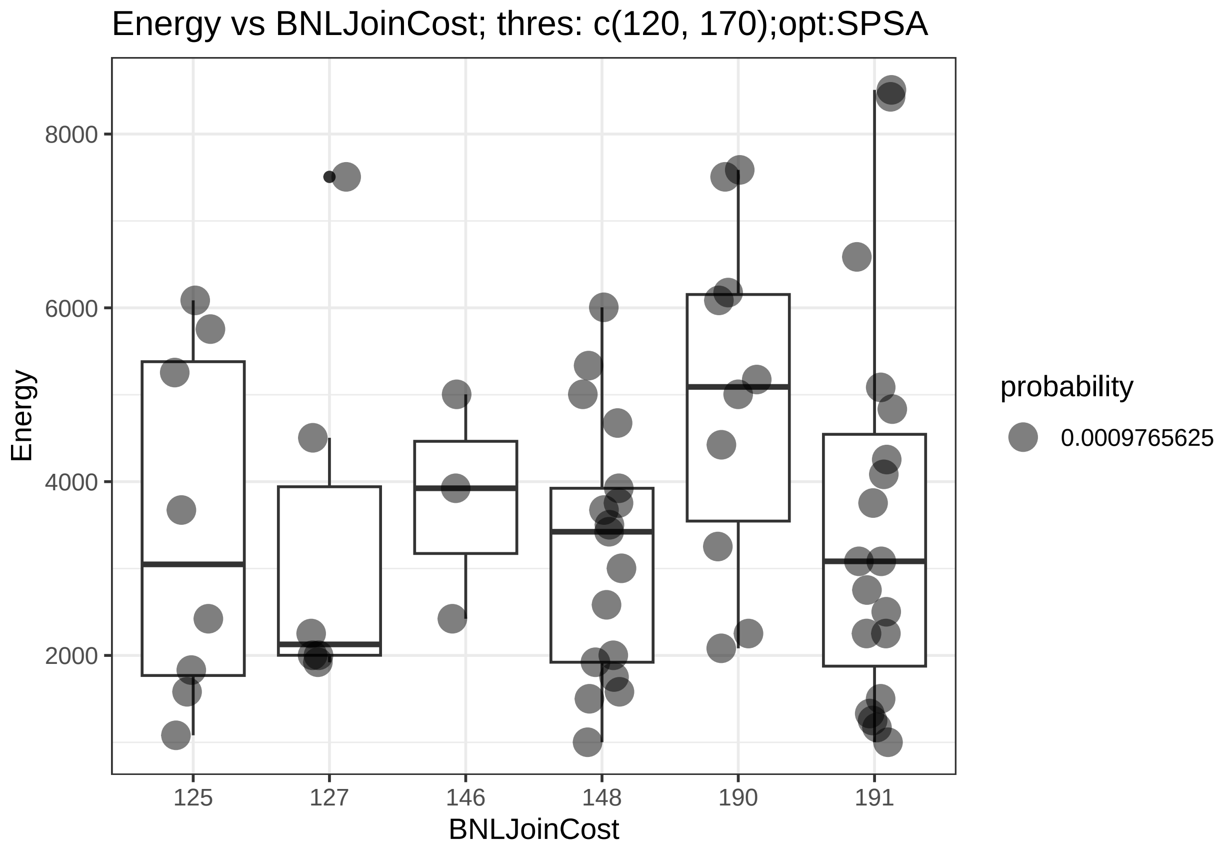


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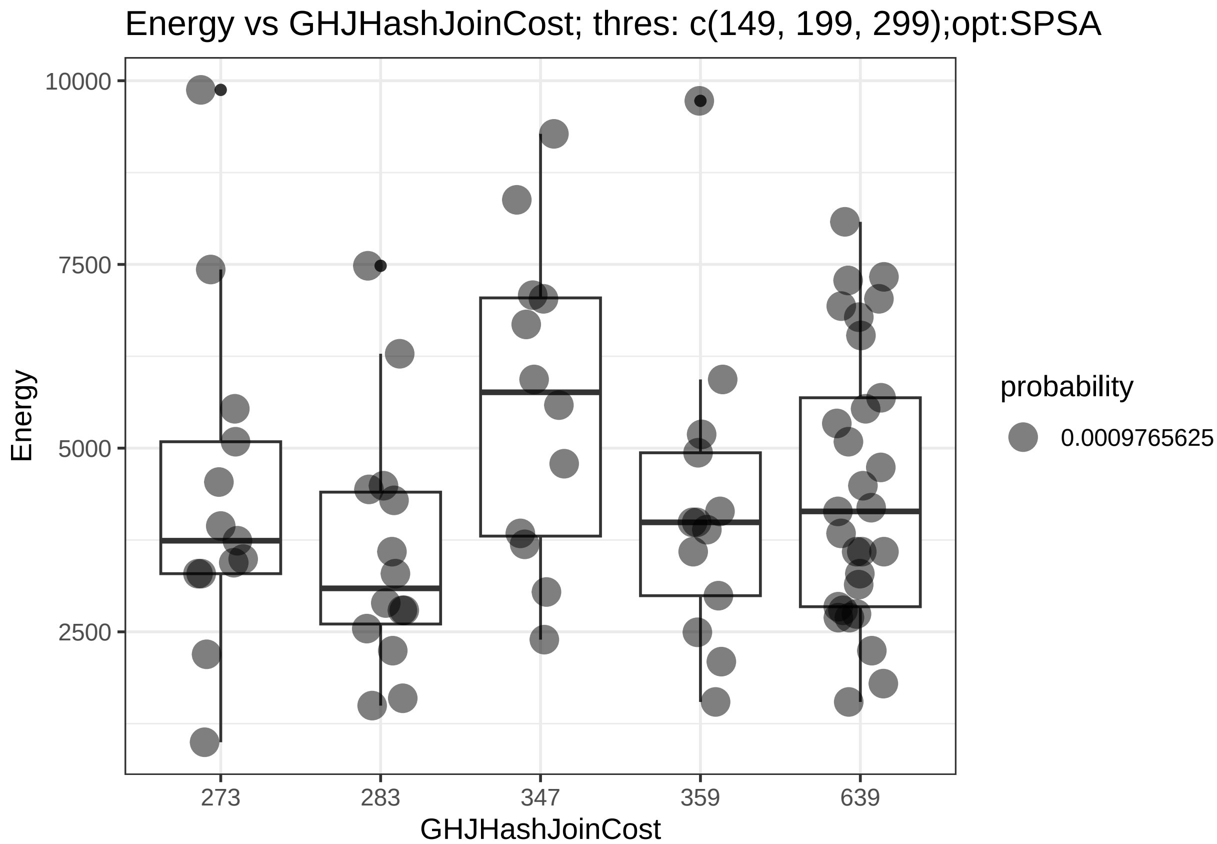


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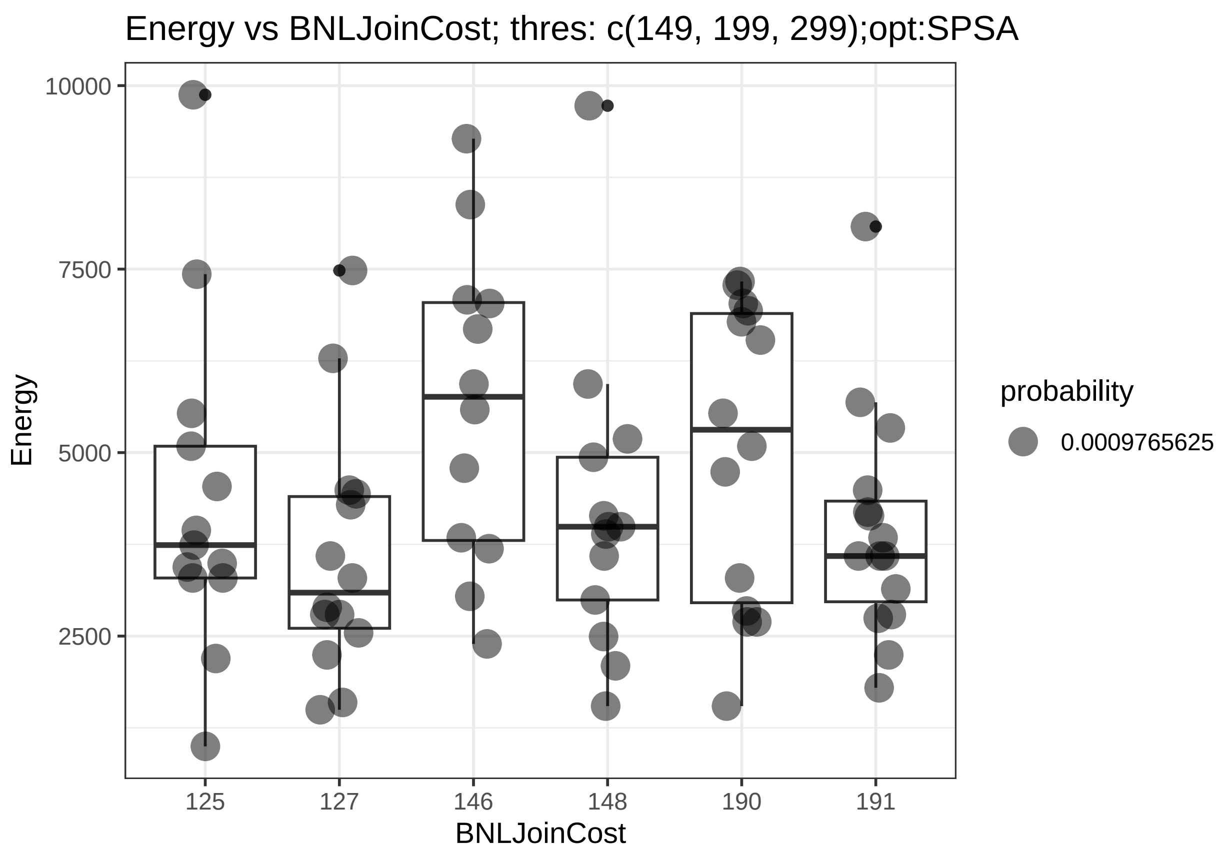
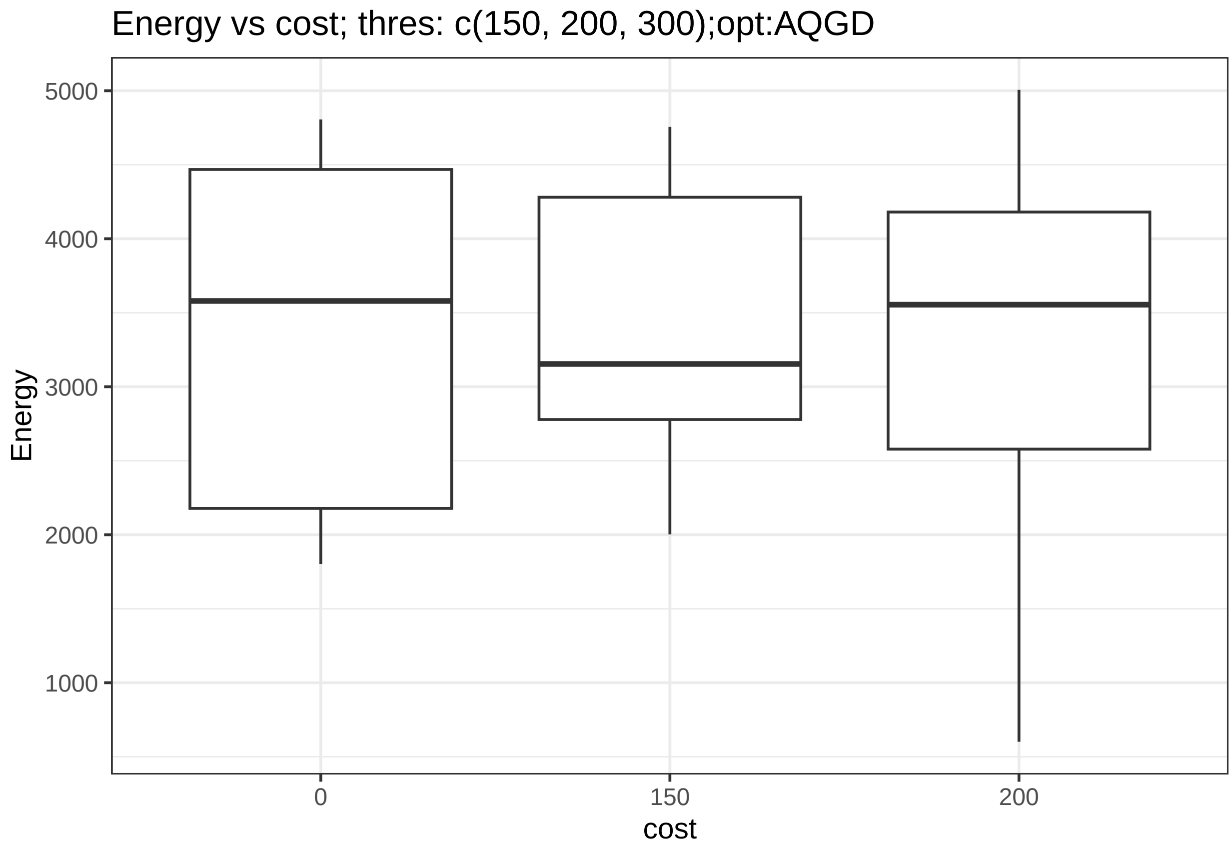



```
## optimizerSPSA; input thres:(c(149, 199, 299)); costType:BNLJoinCost
```

```
inputs=list(c(150, 200, 300),c(160, 200, 240, 280),c(120, 150, 180, 220, 260, 300))
targetCols=c("cost")
for (opt in c("AQGD","COBYLA","SPSA")) {
  for(input in c(0,1,2)){
    for(target in targetCols){
          res <- cost_energy_boxplot_prob_noweight(glue("/Users/ruokun/Desktop/Project/Week11WithI:OCos
    print(glue("optimizer{opt}; input thres:({toString(inputs[input+1])}); costType:{target}"))
    }
  }


}
```

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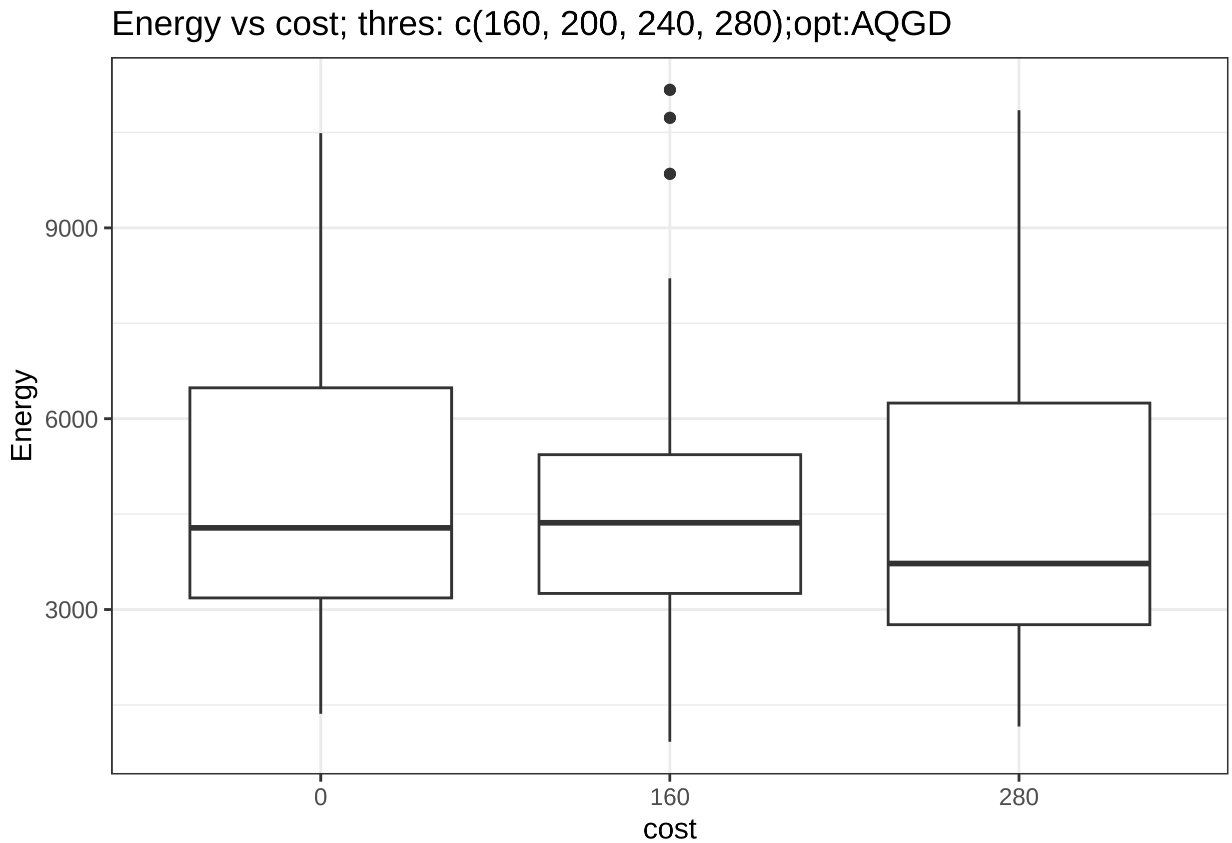
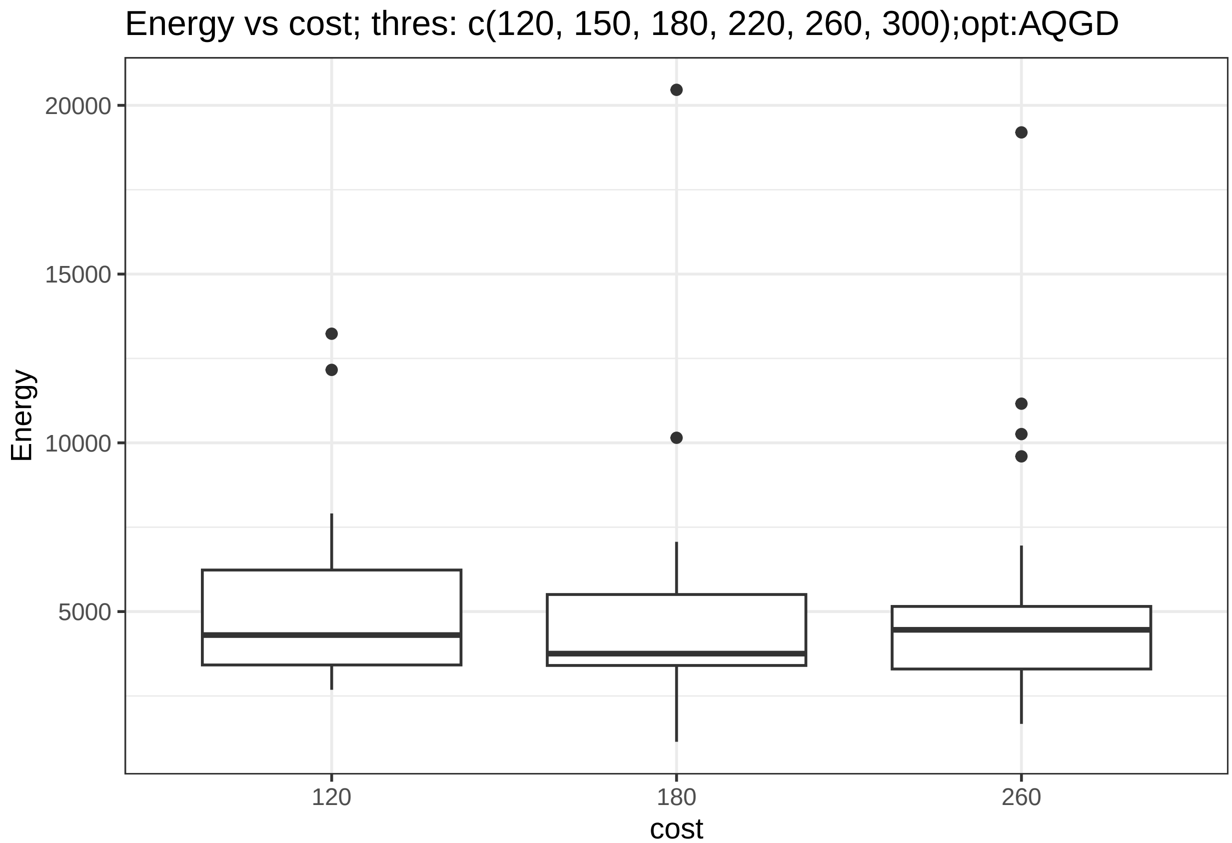



```
## optimizerAQGD; input thres:(c(160, 200, 240, 280)); costType:cost
```

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## optimizerAQGD; input thres:(c(120, 150, 180, 220, 260, 300)); costType:cost
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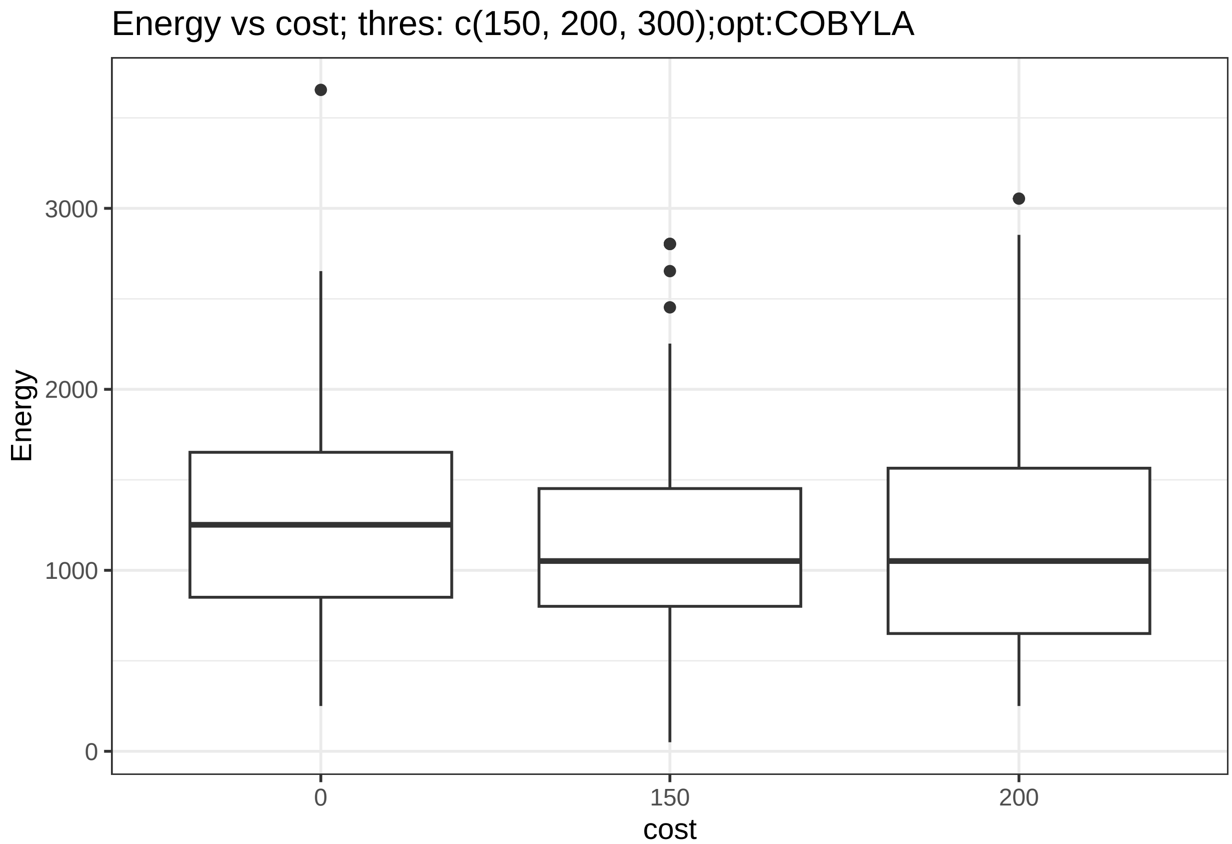


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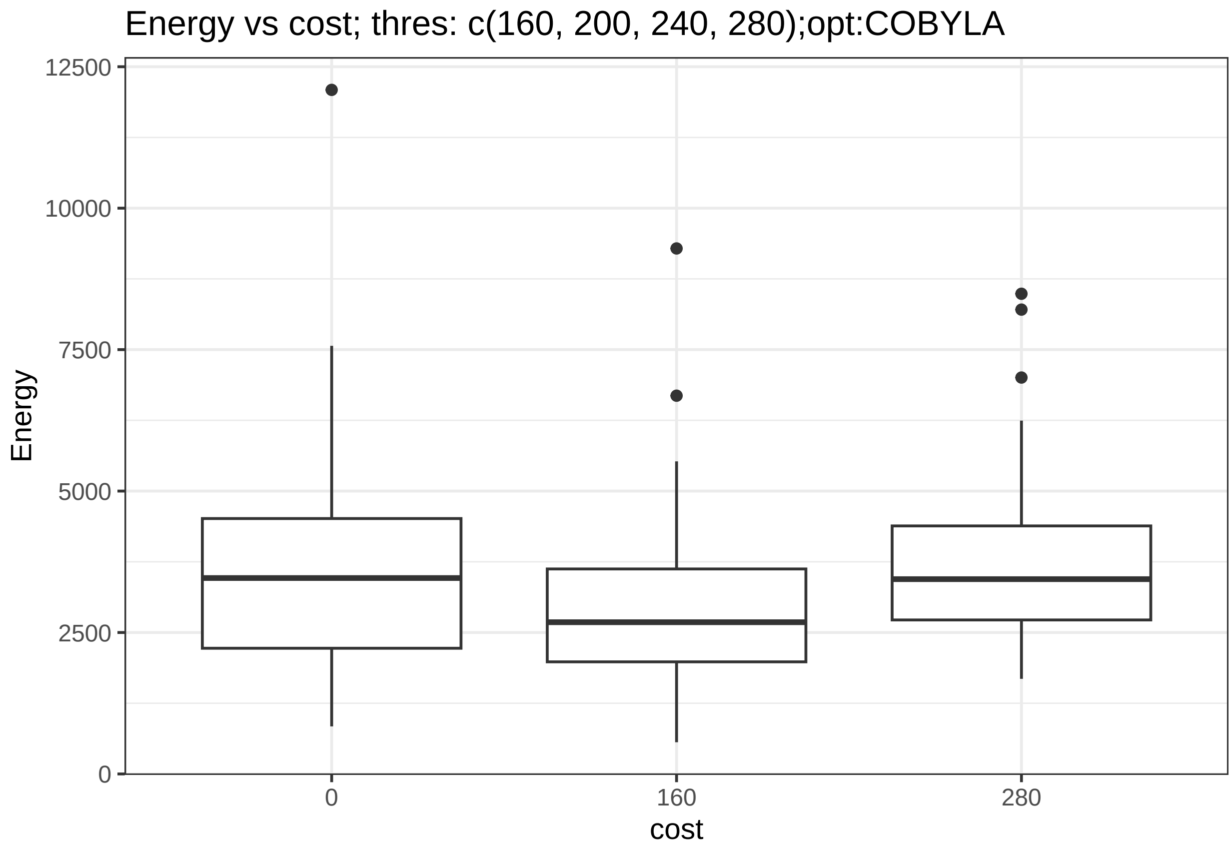


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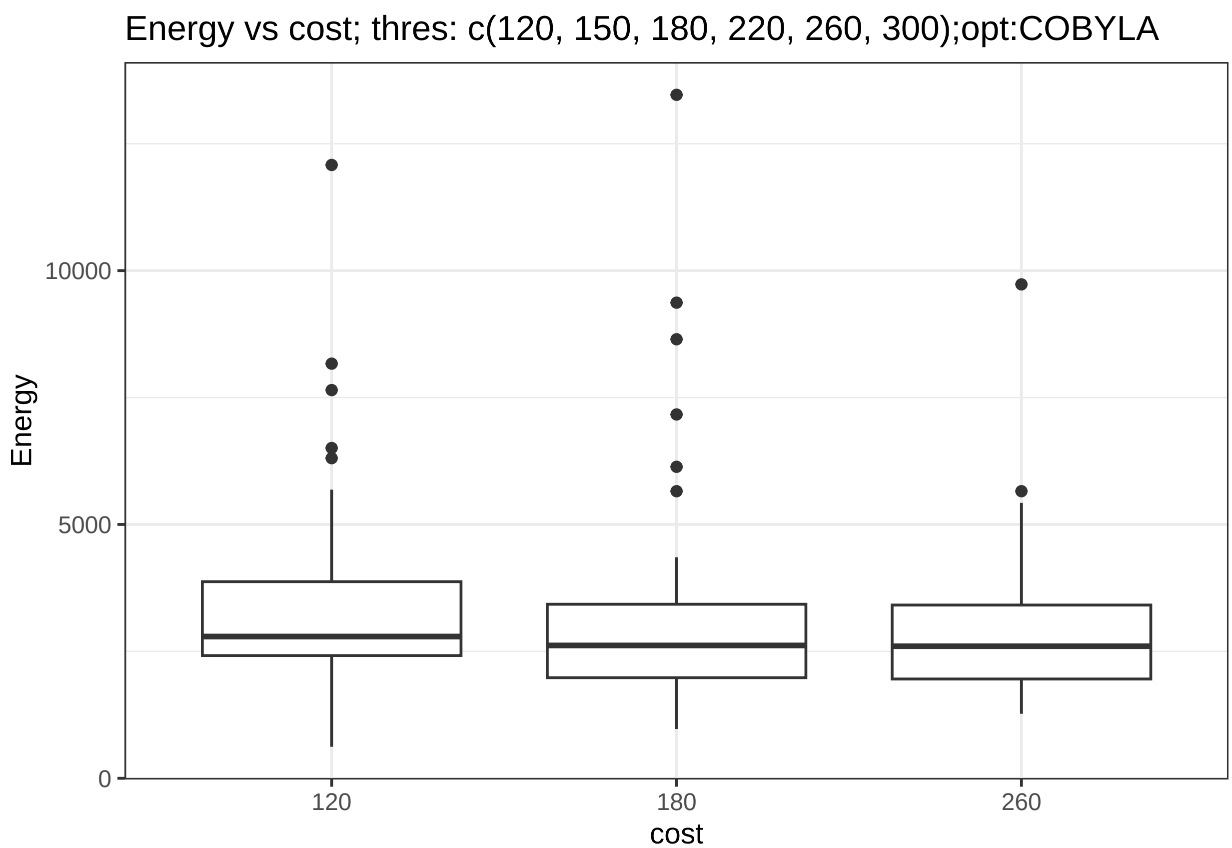


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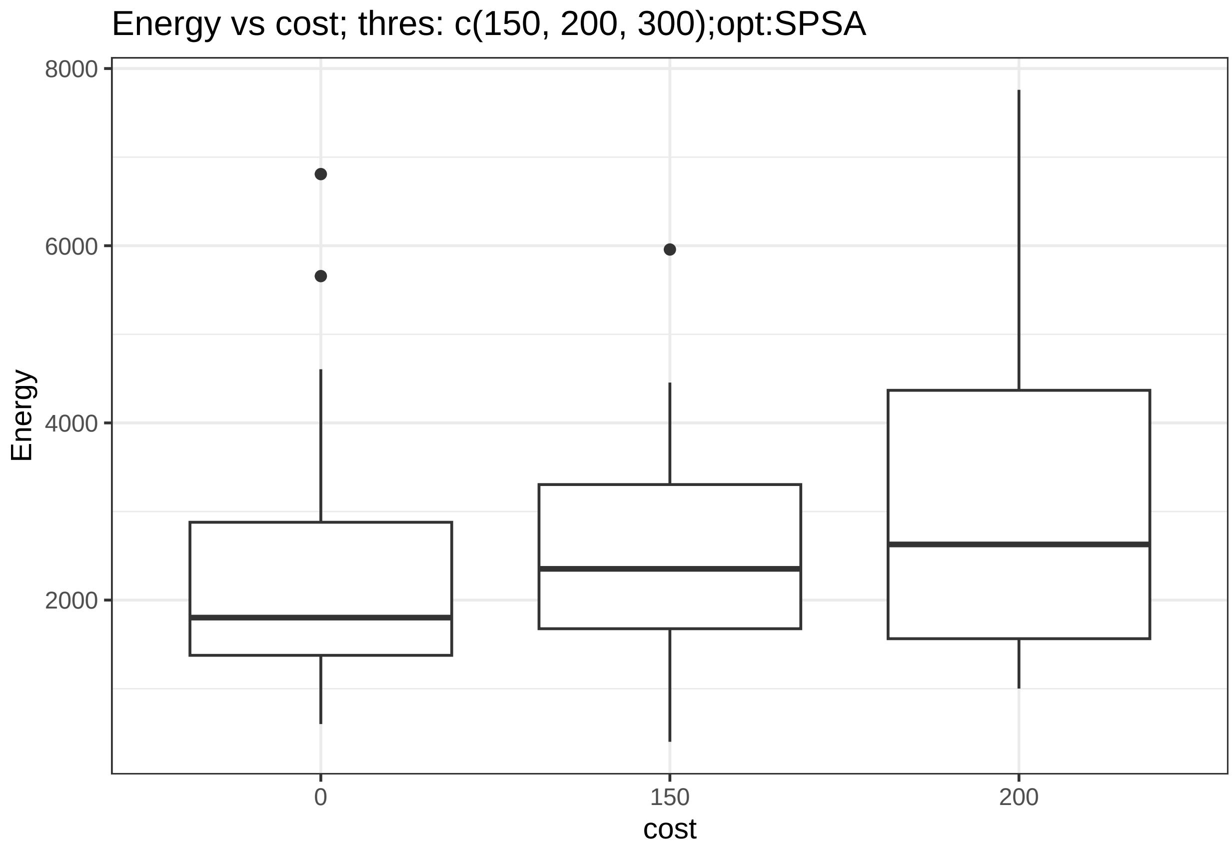


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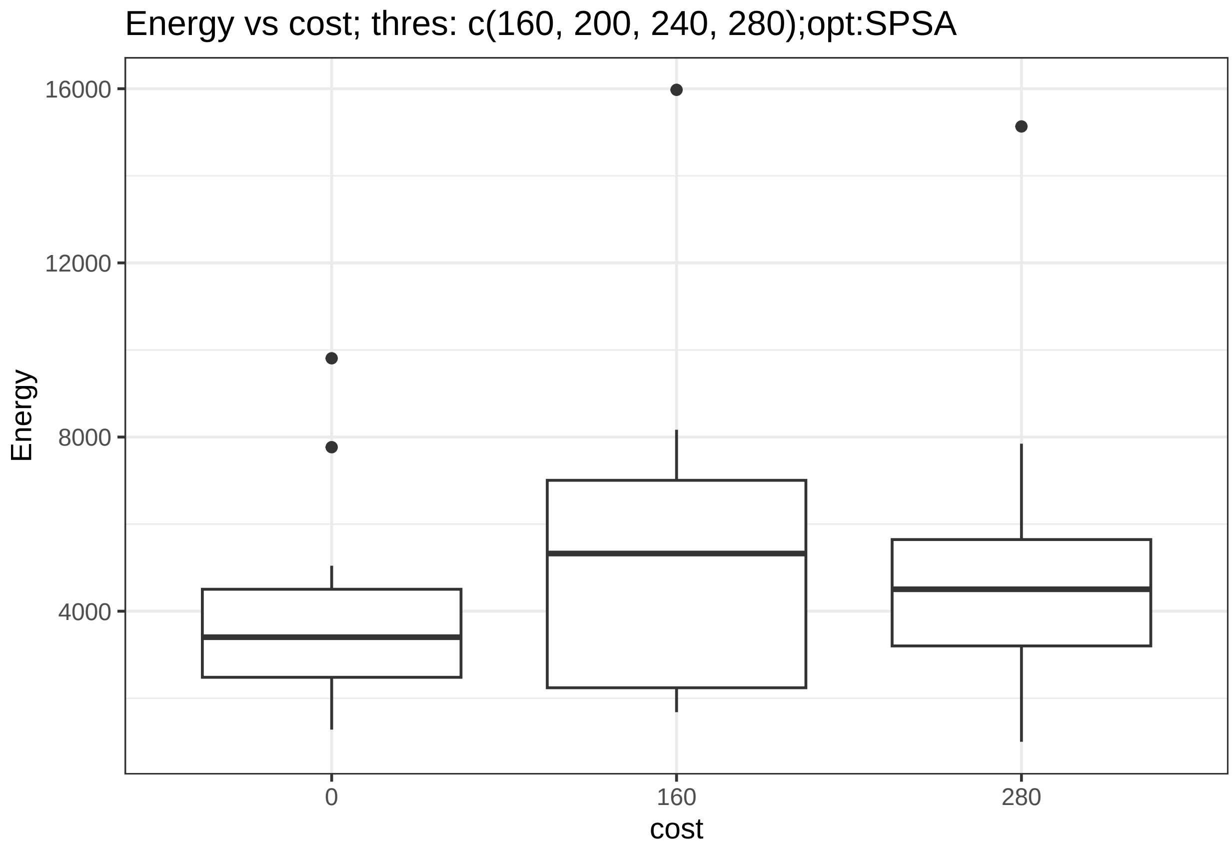


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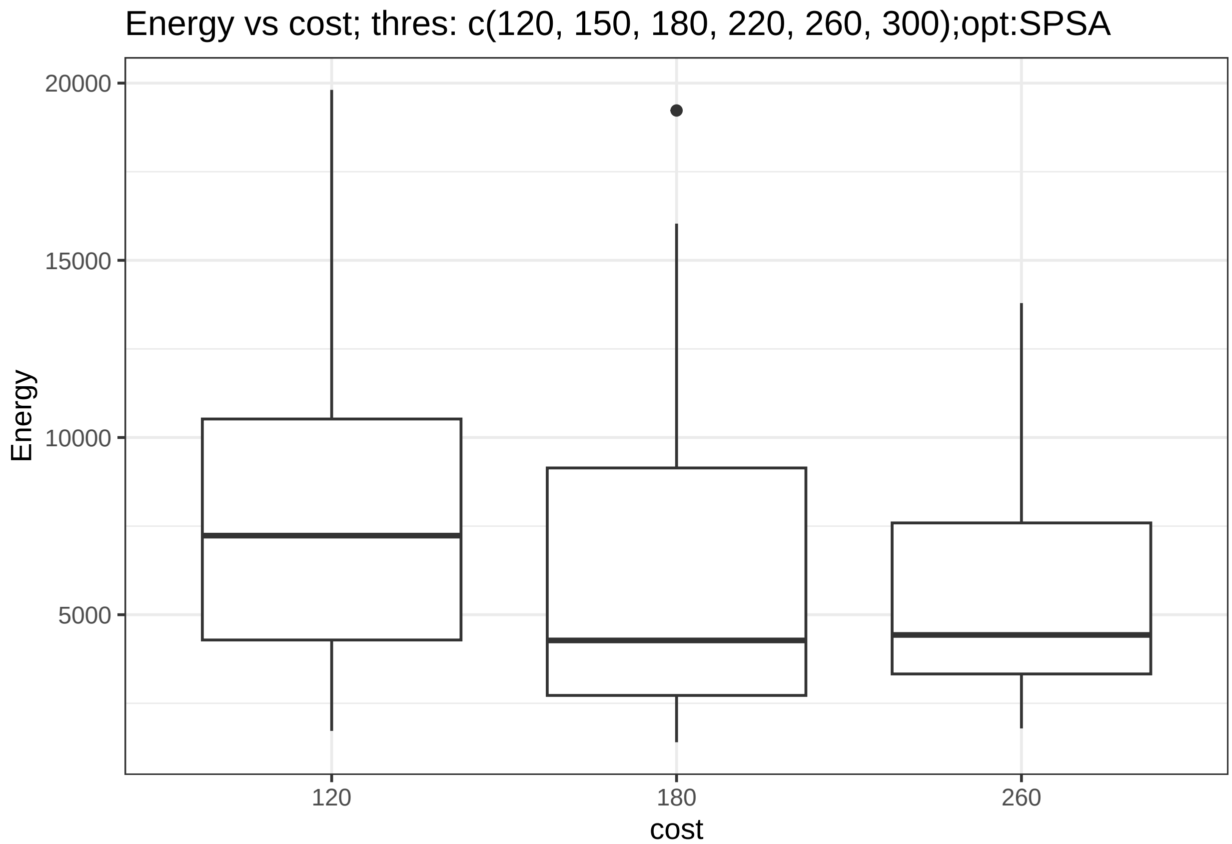


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